8. Numerical methods

What students need to learn:

Location of roots of $f(x) = 0$ by considering changes of sign of $f(x)$ in an interval of x in which $f(x)$ is continuous.	
Approximate solution of equations using simple iterative methods, including recurrence relations of the form $x_{n+1} = f(x_n)$.	Solution of equations by use of iterative procedures for which leads will be given.
Numerical integration of functions.	Application of the trapezium rule to functions covered in C34. Use of increasing number of trapezia to improve accuracy and estimate error will be required. Questions will not require more than three iterations. Simpson's Rule is *not* required.